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YOLOv4 Object Detection Algorithm with Efficient Channel Attention Mechanism

Cui Gao
National Engineering Laboratory for Agri-product Quality Traceability, Beijing Technology and Business University
Beijing Key Laboratory of Big Data Technology, BTBU
Beijing Technology and Business University
Beijing, China
gaocui0825@foxmail.com

Qiang Cai*
National Engineering Laboratory for Agri-product Quality Traceability, Beijing Technology and Business University
Beijing Key Laboratory of Big Data Technology, BTBU
Beijing Technology and Business University
Beijing, China
* Corresponding author: caiq@th.btbu.edu.cn

Shaofeng Ming
National Engineering Laboratory for Agri-product Quality Traceability, Beijing Technology and Business University
Beijing Key Laboratory of Big Data Technology, BTBU
Beijing Technology and Business University
Beijing, China
mingsfeng@foxmail.com

***Abstract*—Channel attention mechanism has been widely used in object detection algorithms because of its strong feature representation ability. The real-time object detection algorithm YOLOv4 has fast detection speed and high accuracy, but it still has some shortcomings, such as inaccurate bounding box positioning and poor robustness. Therefore, we introduced channel attention mechanism into the YOLOv4 algorithm to enhance the feature representation ability of images, and proposed a object detection algorithm with channel attention mechanism. This module firstly carries out global average pooling operation on the features extracted by YOLOv4, and then carries out local cross-channel interactive operation on the feature channels through one-dimensional convolution to enhance the correlation between the features of channels, so as to improve the positioning accuracy of YOLOv4. Our method has achieved good results in the PASCAL VOC dataset. Compared with the original YOLOv4 algorithm, the mAP of this algorithm in the PASCAL VOC test set is improved by 0.62%.**

Keywords: object detection; attention mechanism; deep learning

I. INTRODUCTION

Object detection algorithm based on deep learning is one of the most widely used and challenging tasks in the field of computer vision. It is widely used in many fields such as medical image, unmanned vehicle, security system and robot research. The object detection task is to distinguish the target object in the image from the background information. It usually consists of two parts, namely locating and marking the bounding box of the object to be detected in the image and completing the task of classifying the target in the bounding box. Object detection algorithms based on deep learning can be roughly divided into two categories: two-stage object detection algorithm and one-stage object detection algorithm.

The two-stage object detection algorithm firstly uses the region selection method to generate object area suggestion boxes, and then extracts the characteristics of these suggestions and inputs them into the detection subnetwork to judge the categories of the objects contained therein and correct the coordinates of the bounding boxes. The classical two-stage object detection algorithms include Rich Convolutional Neural Network (R-CNN)[1], Spacial Pyramid Pooling (SPP)[2], Fast-RCNN[3],Faster-RCNN[4], feature Pyramid Network (FPN)[5] and Deformable convolutional networks (DCN)[6] etc. Two-stage object detection algorithm has high accuracy, but its slow detection speed cannot be applied to real life. In recent years, the one-stage object detection algorithm has been widely concerned because of its good real-time performance. The algorithm adopts the end-to-end regression strategy to detect the object in the image, and omits the generation stage of the region suggestion box, thus obtaining the real-time detection speed. The most representative one-stage detection algorithms include You Only Look Once (YOLO [7], YOLOv2[8], YOLOv3[9]) series and Single Shot Multi-Box Detector (SSD) [10]. In order to apply the object detection algorithm to more detection scenes that require real-time performance, researchers improve the accuracy of the one-stage object detection algorithm through different methods.

In recent years, channel attention mechanism has been widely used in one-stage object detection algorithms to improve detection accuracy due to its strong feature representation ability. For example, there are attention-YOLO [11], SSD target detection algorithm based on Attention mechanism and feature fusion [12] etc. All of these algorithms are one-stage, and they can better obtain important features in the image by introducing channel attention mechanism. However, due to the insufficient detection accuracy of these algorithms, despite the addition of channel attention module, a competitive accuracy cannot be achieved. The latest algorithm YOLOv4[13] proposed this year achieves good detection accuracy and speed by improving backbone network and introducing various skills to improve detection accuracy and efficiency.

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However, YOLOv4 extracts the residual network part of the feature and treats each region in the entire feature map equally, considering that each region contributes the same to the final detection. However, in real life scenes, complex and rich contextual information often exists around the object to be detected in the image, and each region in the feature map is treated equally, resulting in a lack of network feature expression ability, inaccurate bounding box positioning, poor robustness and other problems. To solve these problems, we introduced a channel attention mechanism module into the YOLOv4 object detection algorithm. By weighting the features of the object area, we enhanced the expression of useful features for the detection task and inhibited the invalid features to improve the detection accuracy.

II. RELATED WORK

A. YOLOv4 algorithm

YOLOv4[13] algorithm is improved on the basis of YOLOv3[9] algorithm. Its goal is to improve the running speed and optimize parallel computing, so that only a single conventional GPU can be used to train an object detector with good real-time performance and high accuracy. The algorithm consists the heads of backbone network CSPDarknet53[14], space pyramid pooling SPP[2], path aggregation network PANet[15] and YOLOv3[9], and then introduced two pervasive skills, bag-of-Freebies (BoF) and Bag-of-Specials (BoS), to improve the accuracy of subsequent object detection and optimize the parallel computation.

B. Attention mechanism

In recent years, attention mechanism has made important breakthroughs in image, natural language processing and other fields, which have been proved to be beneficial to improve the performance of models. The attention mechanism is essentially similar to the selective visual attention mechanism in humans, the information that is more critical to the current task goal is selected from a large number of information. The function of attention mechanism can be divided into two parts :(1) to determine the input information of the part that needs to be paid attention to;(2) Allocate limited information processing resources to important parts. In the Convolutional Neural Network (CNN), each channel represents a dimension characteristic, so channel-based Attention is also a common mechanism. The essence of channel attention mechanism is to model the importance among various features. For different tasks, features can be assigned according to the input, so that the network can selectively enhance features with large amount of information, so that the subsequent processing can make full use of these features and suppress useless features. In addition, the channel of each feature can be regarded as a class-specific response. The association between channels can be modeled through the channel attention module, and the interdependence relationship between channel graphs can be mined to highlight the interdependent feature graph and enhance the response ability of specific semantic features.

III. YOLOV4 TARGET DETECTION BASED ON CHANNEL ATTENTION MECHANISM

Due to the problems of the YOLO4 object detection algorithm such as inaccurate bounding box positioning and poor robustness, this study will improve the algorithm by inserting channel attention mechanism into the residual module of the CSPDarknet53 backbone network of YOLOv4.

A. YOLOv4 network structure

YOLOv4 searched for the optimal balance between the resolution of the input network, the number of convolutional layers, the number of parameters and the number of output layers through experimental comparison, and selected the CSPDarknet53[14] feature extraction network with low number of parameters and easy training as the backbone. However, the backbone network uses the residual module to extract features, but there is still a lot of background noise information in the extracted features, which will weaken the classification performance of the network in the later stage.

B. Attention-Yolov4 target detection algorithm

YOLOv4[13] algorithm uses full convolutional network structure, small convolution kernel size and end-to-end regression bounding box algorithm to improve the detection speed, but its accuracy is still a certain distance compared with the two-stage. For this reason, this paper proposes an attention-Yolov4[11] object detection algorithm, which can bring high detection accuracy while not increasing network depth and complex. The algorithm further enhances the potential mapping relationship between features by cross-channel interaction of all features extracted from residual modules in the network, which is conducive to reducing training loss and improving the accuracy of positioning and classification. The model structure diagram of the channel attention mechanism Efficient Channel Attention Network (ECANet)[16] is shown in Fig. 1 .

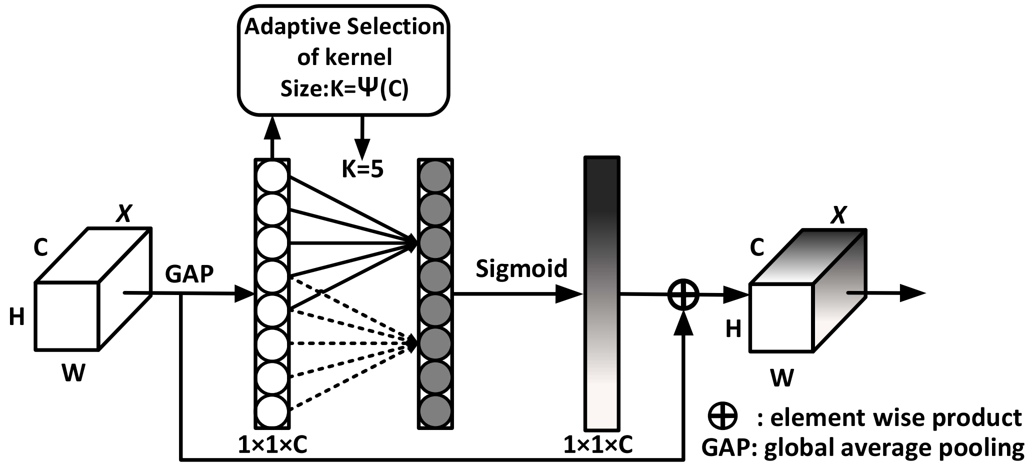

Figure 1. ECANet module structure

In Fig. 1, parameter K is the size of the self-selected one-dimensional convolution kernel, and cross-channel interaction is realized through one-dimensional convolution. The activation function Sigmoid was used to obtain the weight of each channel, and then the expression of useful features in the feature graph was enhanced and invalid features were suppressed by feature weighting.

In this study, after all residual modules of CSPDarknet53[14] feature extraction network, channel attention module ECANet[16] was added. This module is mainly composed of two parts, namely dimensionless local cross-channel interaction and one-dimensional convolution operation with the size of adaptive convolution kernel.

Cross-channel interaction is a new feature combination method that can enhance the feature expression of specific semantics. After global average pooling of the feature map, the ECANet[16] module captures local cross-channel interactions by considering the interactions between each channel feature and k neighboring channels. Cross-channel interaction can avoid dimensionality reduction through one dimensional convolution operation and realize multi-channel interaction effectively. Therefore, the weight of its feature y_i can be calculated as:

$$w_i = \sigma\left(\sum_{j=1}^{k} \alpha^j y_i^j\right), y_i^j \in \Omega_i^k \tag{1}$$

Where, σ represents the Sigmoid function, Ω_i^k represents the set of k adjacent channels of y_i, and formula (1) represents that all channels share the same inclination parameter, thus allowing higher model efficiency.

$$\mathrm{w} = \sigma\big(\mathrm{C1D}_k(y)\big) \tag{2}$$

Formula (2) represents the realization of channel feature cross-channel interaction with one-dimensional convolution(C1D), in which C1D represents one-dimensional convolution and k represents the coverage of local cross-channel interaction, how many neighbors near the channel participate in the attention prediction of this channel.

In order to avoid manual tuning k through cross-validation, the ECANet[16] module proposes an adaptive method to determine the value size. Where, the interaction coverage (i.e., the convolution kernel size k) is proportional to the channel dimension C, and the two are nonlinear mapping relations, as shown in Formula (3).

$$\mathrm{C} = \phi(k) = 2^{\,(\gamma * k - b)} \tag{3}$$

Given the channel dimension C, the convolution kernel size K can be adaptively determined, as shown in formula (4).

$$\mathrm{k} = \psi(\mathrm{C}) = \left|\frac{\log_2(C)}{\gamma} + \frac{b}{\gamma}\right|_{odd} \tag{4}$$

Where t is closest to $|t|_{odd}$, Set γ and b to 2 and 1, respectively, for the purposes of this article. Through formula (4), k adjacent channels can be reasonably selected for local information interaction.

In this experiment, attention mechanism ECANet[16] was added to the residual module of YOLOv4's CSPDarknet[14] backbone network, and the residual module with attention mechanism was called ECA-ResNet. This module is mainly composed of three parts, namely global average pooling process, cross-channel interaction (one-dimensional convolution) and feature weighting operation. The structure of ECA-ResNet module is shown in Fig. 2.

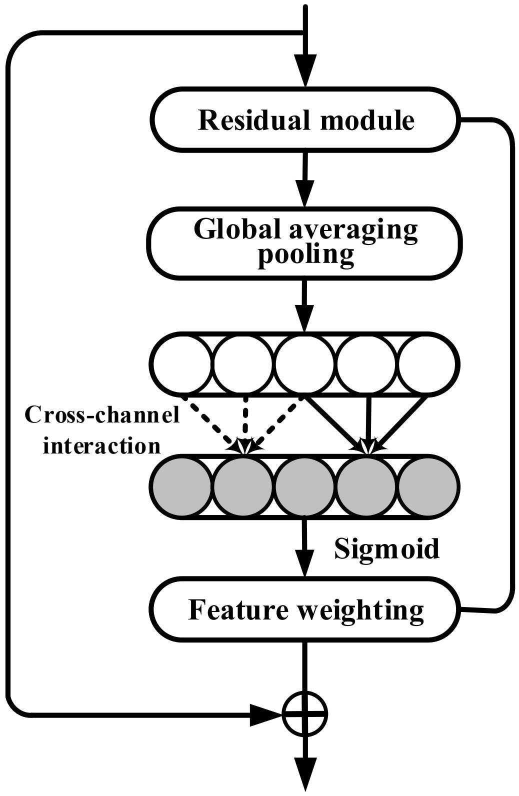

Figure 2. Structure diagram of ECA-ResNet module

In Fig. 2, ECA-Resnet first input the feature graph extracted from the residual module into the global average pooling layer to obtain the feature value of channel information of each dimension. Then, the one-dimensional convolution with the size of the adaptive convolution kernel is used for dimensionless inter-feature cross-channel interaction operation. Finally, the weights of features in different regions are added, and the weighted feature map is transferred to the rest of the network. In this module, the value of local cross-channel interactive coverage is determined by the method of adaptive convolution kernel size selection, which avoids the need to tune the value through cross-validation. ECANet[16] module based on the correlation of the channel is modeled in order to improve the network's ability to represent characteristics, and channel features according to how important they are assigned different weights, so network can learn the characteristics of the global information choice to strengthen useful information and suppress useless information, thus improve the image localization and classification accuracy.

IV. Experimental results and analysis

The data set used in the experiment was the PASCAL VOC dataset[17], which consisted of a total of 27,088 images for 20 categories. In this paper, PASCAL VOC 2007 and PASCAL VOC 2012 datasets were used for training, and PASCAL VOC 2007 test sets were used for testing. Based on pytorch framework, image processor (GPU) was used for accelerated operation. The experimental platform was Intel(R) Xeon(R) CPU E5-2603 V3@1.60ghz processor, 8.00 GB memory, and Nvidia Tesla K40M graphics card.

A. Analysis of experimental results

In object detection, mAP index is usually used to evaluate the accuracy. Based on the original YOLOv4 model, the resolution of the input image was set at 416×416, the Batch size was 8, the initial learning rate was 0.0001, and the final network model was obtained after 50 iterations. Compared with the original YOLOv4 algorithm, SE-Yolov4 algorithm, and the 20 categories of this paper's ECA-Yolov4 algorithm, the experimental results are shown in TABLE I.

TABLE II. compares the three kinds of the change of the accuracy and real-time performance of the algorithm, put together the mAP of each algorithm and the detection speed, this paper introduced the attention mechanism of ECANet module YOLOv4 compared to the original algorithm, the precision is improved by 0.62%, real time by a slight decline, but the ECA -YOLOv4 algorithm compared with SE - YOLOv4 detection speed a lot, this is due to the ECANet module into fewer, so the original YOLOv4 detection speed of the algorithm.

TABLE I .COMPARISON OF DETECTION ACCURACY OF 3 OBJECT DETECTION ALGORITHMS UNDER 20 CATEGORIE

algorithm / **category**	*YOLOv4*	*SE-YOLOv4*	*ECA-YOLOv4(ours)*
aeroplane	**71.5**	68.2	70.5
bicycle	76.1	74.3	**77.8**
bird	55.6	**60.6**	57.6
boat	**55.2**	48.2	54.6
bottle	43.8	31.1	**51.3**
bus	**75.6**	71.1	75.0
car	85.2	75.8	**86.7**
cat	71.4	**79.7**	70.0
chair	45.9	46.8	49.2
cow	60.3	**66.4**	62.2
diningtable	64.7	**65.5**	65.1
dog	65.1	**75.7**	62.8
horse	78.0	**81.9**	77.7
motorbike	76.1	77.5	**81.7**
person	**81.6**	72.1	81.3
pottedplant	40.5	39.7	**43.4**
sheep	63.0	61.9	**63.8**
sofa	63.8	**65.7**	61.9
train	74.6	**76.6**	74.0
tvmonitor	68.2	62.4	**68.6**
mean	62.0	65.0	**66.78**

B. Comparison of experimental results

At the same time, one-dimensional convolution is used to carry out cross-channel interaction on channel features to enhance the semantic information of the high-level feature map. The improved algorithm can better locate the characteristics of the object to be detected, improve the generalization performance of the network without introducing too many parameters, and improve the accuracy of the algorithm while reducing the complexity of the model. The experimental results showed in Fig. 3 that the YOLOv4 object detection algorithm based on channel attention mechanism proposed in this paper achieves better detection accuracy than the original YOLOv4 algorithm under the premise of introducing very few parameters and hardly increasing the computational complexity.

TABLE II.THREE TARGET DETECTION ALGORITHMS ON THE PASCAL VOC 2007 TEST SET COMPARISON OF DETECTION SPEED

algorithm	***mAP/%***	***Detection speed /(frame·s^{-1})***
YOLOv4	65.40	75.64 ms
SE-YOLOv4	65.93	137.76 ms
ECA-YOLOv4(ours)	66.02	81.40 ms

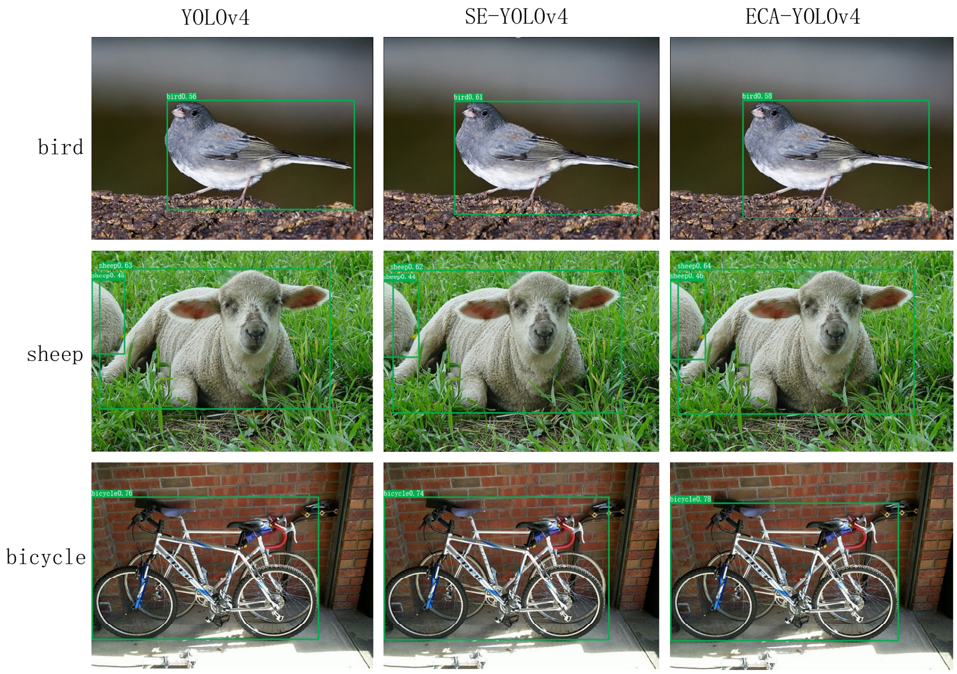

Figure 3. Comparison diagram of experimental result

V. Experimental Results and Analysis

In this paper, an improved YOLOv4 object detection algorithm, attention-Yolov4, is proposed with the efficient channel Attention mechanism. This algorithm can improve the accuracy at a relatively low additional computational cost. In this paper, the algorithm introduced channel attention mechanism ECANet into the residual module of YOLOv4's backbone network, CSPDarknet53, and added different weights to the channel features to make the network pay more attention to the important features containing the object to be detected, while suppressing the invalid features affecting the final detection results, thus enhancing the robustness of the algorithm. Through experimental comparison, the detection accuracy of this algorithm has been improved to some extent. In the future, we will gradually improve the model and further improve the detection accuracy of this algorithm on the basis of considering the improvement of high-level features.

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